

Лабораторная работа №8

Настройка SMTP-сервера (Postfix)

Владимир Базлов

29 ноября 2025

Российский университет дружбы народов

Цель работы

Овладеть установкой и настройкой почтового сервера **Postfix**,
а также диагностикой доставки локальных и доменных сообщений.

Выполнение работы

Установка Postfix и s-nail

```
Installed:
  postfix-2:3.8.5-8.el10.x86_64          postfix-ldap-2:3.8.5-8.el10.x86_64          s-nail-14.9.24-12.el10.x86_64

Complete!
[root@server.vabazlov.net ~]#
[root@server.vabazlov.net ~]# firewall-cmd --add-service=smtp
success
[root@server.vabazlov.net ~]# firewall-cmd --add-service=smtp --permanent
success
[root@server.vabazlov.net ~]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http https smtp ssh ssh-custom
[root@server.vabazlov.net ~]# restorecon -vR /etc
Relabeled /etc/NetworkManager/system-connections/eth1.nmconnection from unconfined_u:object_r:user_tmp_t:s0 to unconfined_u:object_r:NetworkManager_etc_rw_t:s0
[root@server.vabazlov.net ~]# systemctl enable postfix.service
Created symlink '/etc/systemd/system/multi-user.target.wants/postfix.service' → '/usr/lib/systemd/system/postfix.service'.
[root@server.vabazlov.net ~]# systemctl start postfix.service
[root@server.vabazlov.net ~]# █
```

Рис. 1: Установка postfix и s-nail

Проверка списка параметров Postfix

```
virtual_mailbox_base =  
virtual_mailbox_domains = $virtual_mailbox_maps  
virtual_mailbox_limit = 51200000  
virtual_mailbox_lock = fcntl, dotlock  
virtual_mailbox_maps =  
virtual_minimum_delivery_slots = $default_minimum_delivery_slots  
virtual_minimum_uid = 100  
virtual_recipient_limit = $default_recipient_limit  
virtual_recipient_refill_delay = $default_recipient_refill_delay  
virtual_recipient_refill_limit = $default_recipient_refill_limit  
virtual_transport = virtual  
virtual_transport_rate_delay = $default_transport_rate_delay  
virtual_uid_maps =  
[root@server.vabazlov.net ~]#  
[root@server.vabazlov.net ~]# postconf myorigin  
myorigin = $myhostname  
[root@server.vabazlov.net ~]# postconf mydomain  
mydomain = vabazlov.net  
[root@server.vabazlov.net ~]# postconf -e 'myorigin = $mydomain'  
[root@server.vabazlov.net ~]# postconf myorigin  
myorigin = $mydomain  
[root@server.vabazlov.net ~]# postfix check  
[root@server.vabazlov.net ~]# systemctl reload postfix.service  
[root@server.vabazlov.net ~]#
```

Параметры myorigin и mydomain

```
[root@server.vabazlov.net ~]#  
[root@server.vabazlov.net ~]# postconf -n  
alias_database = ldap:/etc/aliases  
alias_maps = ldap:/etc/aliases  
command_directory = /usr/sbin  
compatibility_level = 3.8  
daemon_directory = /usr/libexec/postfix  
data_directory = /var/lib/postfix  
debug_peer_level = 2  
debugger_command = PATH=/bin:/usr/bin:/usr/local/bin:/usr/X11R6/bin ddd $daemon_directory/$process_name $process_id & sleep 5  
default_database_type = ldap  
html_directory = no  
inet_interfaces = localhost  
inet_protocols = all  
mail_owner = postfix  
mailq_path = /usr/bin/mailq.postfix  
manpage_directory = /usr/share/man  
meta_directory = /etc/postfix  
mydestination = $myhostname, localhost.$mydomain, localhost  
myorigin = $mydomain  
newaliases_path = /usr/bin/newaliases.postfix  
queue_directory = /var/spool/postfix  
readme_directory = /usr/share/doc/postfix/README_FILES  
sample_directory = /usr/share/doc/postfix/samples  
sendmail_path = /usr/sbin/sendmail.postfix  
setgid_group = postdrop  
shlib_directory = /usr/lib64/postfix  
smtp_tls_CAfile = /etc/pki/tls/certs/ca-bundle.crt  
smtp_tls_CApith = /etc/pki/tls/certs  
smtp_tls_security_level = may  
smtpd_tls_cert_file = /etc/pki/tls/certs/postfix.pem  
smtpd_tls_key_file = /etc/pki/tls/private/postfix.key  
smtpd_tls_security_level = may  
unknown_local_recipient_reject_code = 550  
[root@server.vabazlov.net ~]# postconf -e 'mydomain = vabazlov.net'  
[root@server.vabazlov.net ~]# postconf inet_protocols  
inet_protocols = all  
[root@server.vabazlov.net ~]# postconf -e 'inet_protocols = ipv4'  
[root@server.vabazlov.net ~]# postfix check  
[root@server.vabazlov.net ~]# systemctl reload postfix.service  
[root@server.vabazlov.net ~]#
```

Вывод postconf -n

```
[vabazlov@server.vabazlov.net ~]$ echo . | mail -s terst1 vabazlov@server.vabazlov.net
[vabazlov@server.vabazlov.net ~]$
[vabazlov@server.vabazlov.net ~]$
[vabazlov@server.vabazlov.net ~]$ ls /var/spool/mail/
vabazlov  vagrant
[vabazlov@server.vabazlov.net ~]$ cat /var/spool/mail/vabazlov
From vabazlov@vabazlov.net  Sat Nov 29 07:34:16 2025
Return-Path: <vabazlov@vabazlov.net>
X-Original-To: vabazlov@server.vabazlov.net
Delivered-To: vabazlov@server.vabazlov.net
Received: by server.vabazlov.net (Postfix, from userid 1001)
        id B0E9F19B94B; Sat, 29 Nov 2025 07:34:16 +0000 (UTC)
Date: Sat, 29 Nov 2025 07:34:16 +0000
To: vabazlov@server.vabazlov.net
Subject: terst1
User-Agent: s-nail v14.9.24
Message-Id: <20251129073416.B0E9F19B94B@server.vabazlov.net>
From: vabazlov@vabazlov.net

.

[vabazlov@server.vabazlov.net ~]$
[vabazlov@server.vabazlov.net ~]$
```

Рис. 4: Вывод postconf -n


```
Nov 29 07:34:16 server postfix/pickup[12193]: B0E9F19B94B: uid=1001 from=<vabazlov>  
Nov 29 07:34:16 server postfix/cleanup[12662]: B0E9F19B94B: message-id=<20251129073416.B0E9F19B94B@server.vabazlov.net>  
Nov 29 07:34:16 server postfix/qmgr[12194]: B0E9F19B94B: from=<vabazlov@vabazlov.net>, size=333, nrcpt=1 (queue active)  
Nov 29 07:34:16 server postfix/local[12665]: B0E9F19B94B: to=<vabazlov@server.vabazlov.net>, relay=local, delay=0.02, delays=0.01/0/0/0, dsn=2.0.  
0, status=sent (delivered to mailbox)  
Nov 29 07:34:16 server postfix/qmgr[12194]: B0E9F19B94B: removed
```

Рис. 5: Полученное письмо на сервере

```
Installed:
 postfix-2:3.8.5-8.el10.x86_64      postfix-lmdb-2:3.8.5-8.el10.x86_64
 s-nail-14.9.24-12.el10.x86_64

Complete!
[root@client.vabazlov.net ~]#
[root@client.vabazlov.net ~]# postconf inet_protocols
inet_protocols = all
[root@client.vabazlov.net ~]# postconf -e 'inet_protocols = ipv4'
[root@client.vabazlov.net ~]# systemctl enable postfix.service
Created symlink '/etc/systemd/system/multi-user.target.wants/postfix.service' →
'/usr/lib/systemd/system/postfix.service'.
[root@client.vabazlov.net ~]# systemctl start postfix.service
[root@client.vabazlov.net ~]# █
```

Рис. 6: Логи доставки локального письма

```
[vabazlov@client.vabazlov.net ~]$  
[vabazlov@client.vabazlov.net ~]$ echo . | mail -s test1 vabazlov@server.vabazlov  
.net  
[vabazlov@client.vabazlov.net ~]$ █
```

Рис. 7: Установка postfix на клиенте

Новые подключения после перезапуска Postfix

```
[root@server.vabazlov.net ~]#  
[root@server.vabazlov.net ~]# postconf inet_interfaces  
inet_interfaces = localhost  
[root@server.vabazlov.net ~]# postconf mynetworks  
mynetworks = 127.0.0.1/32  
[root@server.vabazlov.net ~]# postconf -e 'inet_interfaces = all'  
[root@server.vabazlov.net ~]# postconf -e 'mynetworks = 127.0.0.0/8, 192.168.0.0/16'  
[root@server.vabazlov.net ~]# postfix check  
[root@server.vabazlov.net ~]# systemctl reload postfix.service  
[root@server.vabazlov.net ~]# systemctl stop postfix.service  
[root@server.vabazlov.net ~]# systemctl start postfix.service  
[root@server.vabazlov.net ~]# █
```

Рис. 8: Перезапуск Postfix и новые подключения

```
Nov 29 07:43:23 server postfix/postfix-script[14042]: starting the Postfix mail system
Nov 29 07:43:23 server postfix/master[14044]: daemon started -- version 3.8.5, configuration /etc/postfix
Nov 29 07:43:35 server postfix/smtpd[14078]: connect from client.vabazlov.net[192.168.1.30]
Nov 29 07:43:35 server postfix/smtpd[14078]: 3EE1119CE82: client=client.vabazlov.net[192.168.1.30]
Nov 29 07:43:35 server postfix/cleanup[14082]: 3EE1119CE82: message-id=<20251129074335.2ED4A6134533@client.vabazlov.net>
Nov 29 07:43:35 server postfix/smtpd[14078]: disconnect from client.vabazlov.net[192.168.1.30] ehlo=2 starttls=1 mail=1 rcpt=1 data=1 quit=1 commands=7
Nov 29 07:43:35 server postfix/qmgr[14046]: 3EE1119CE82: from=<vabazlov@client.vabazlov.net>, size=552, nrcpt=1 (queue active)
Nov 29 07:43:35 server postfix/local[14083]: 3EE1119CE82: to=<vabazlov@server.vabazlov.net>, relay=local, delay=0.01, delays=0/0/0/0, dsn=2.0.0, status=sent (delivered to mailbox)
Nov 29 07:43:35 server postfix/qmgr[14046]: 3EE1119CE82: removed
```

Рис. 9: Доставка писъма от клиента

Почтовые файлы с двумя сообщениями

```
[vabazlov@server.vabazlov.net ~]$  
[vabazlov@server.vabazlov.net ~]$ cat /var/spool/mail/vabazlov  
From vabazlov@vabazlov.net Sat Nov 29 07:34:16 2025  
Return-Path: <vabazlov@vabazlov.net>  
X-Original-To: vabazlov@server.vabazlov.net  
Delivered-To: vabazlov@server.vabazlov.net  
Received: by server.vabazlov.net (Postfix, from userid 1001)  
        id B0E9F19B94B; Sat, 29 Nov 2025 07:34:16 +0000 (UTC)  
Date: Sat, 29 Nov 2025 07:34:16 +0000  
To: vabazlov@server.vabazlov.net  
Subject: terst1  
User-Agent: s-nail v14.9.24  
Message-Id: <20251129073416.B0E9F19B94B@server.vabazlov.net>  
From: vabazlov@vabazlov.net  
  
.  
  
From vabazlov@client.vabazlov.net Sat Nov 29 07:43:35 2025  
Return-Path: <vabazlov@client.vabazlov.net>  
X-Original-To: vabazlov@server.vabazlov.net  
Delivered-To: vabazlov@server.vabazlov.net  
Received: from client.vabazlov.net (client.vabazlov.net [192.168.1.30])  
        by server.vabazlov.net (Postfix) with ESMTPS id 3EE1119CE82  
        for <vabazlov@server.vabazlov.net>; Sat, 29 Nov 2025 07:43:35 +0000 (UTC)  
Received: by client.vabazlov.net (Postfix, from userid 1001)  
        id 2ED4A6134533; Sat, 29 Nov 2025 07:43:35 +0000 (UTC)  
Date: Sat, 29 Nov 2025 07:43:35 +0000  
To: vabazlov@server.vabazlov.net  
Subject: test1  
User-Agent: s-nail v14.9.24  
Message-Id: <20251129074335.2ED4A6134533@client.vabazlov.net>  
From: vabazlov@client.vabazlov.net  
  
.  
  
[vabazlov@server.vabazlov.net ~]$
```

Отправка письма на доменный адрес

```
[vabazlov@client.vabazlov.net ~]$  
[vabazlov@client.vabazlov.net ~]$ echo . | mail -s test2 vabazlov@vabazlov.net  
[vabazlov@client.vabazlov.net ~]$ postqueue -p  
-Queue ID- --Size-- ----Arrival Time---- -Sender/Recipient-----  
967D56134530      341 Sat Nov 29 07:41:16 vabazlov@client.vabazlov.net  
                (connect to server.vabazlov.net[192.168.1.1]:25: Connection refused)  
                vabazlov@server.vabazlov.net  
  
-- 0 Kbytes in 1 Request.  
[vabazlov@client.vabazlov.net ~]$
```

Рис. 11: Отправка письма на доменный адрес

Ошибка loops back to myself / No route to host

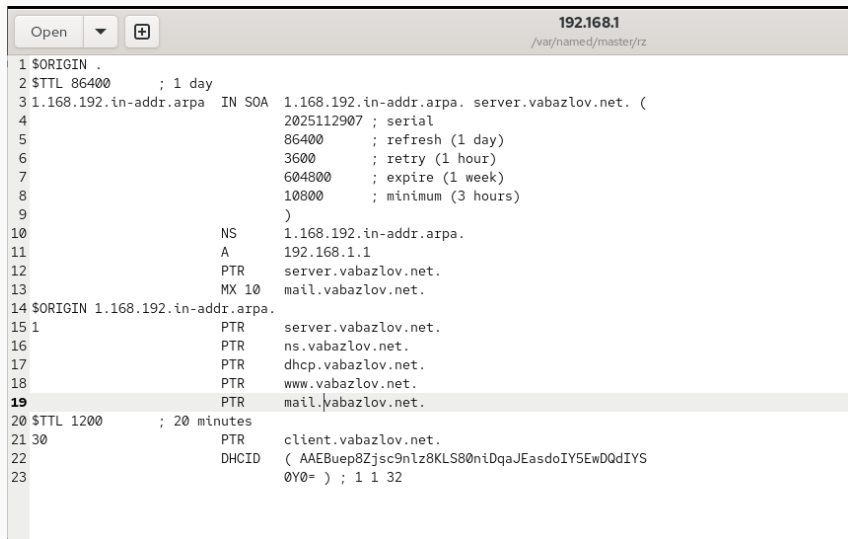
```
Nov 29 07:44:40 server postfix/smtpd[14078]: connect from client.vabazlov.net[192.168.1.30]
Nov 29 07:44:40 server postfix/smtpd[14078]: 0A07E19CE82: client=client.vabazlov.net[192.168.1.30]
Nov 29 07:44:40 server postfix/cleanup[14082]: 0A07E19CE82: message-id=<20251129074440.084126134533@client.vabazlov.net>
Nov 29 07:44:40 server postfix/smtpd[14078]: disconnect from client.vabazlov.net[192.168.1.30] ehlo=2 starttls=1 mail=1 rcpt=1 data=1 quit=1 comm
ands=7
Nov 29 07:44:40 server postfix/qmgr[14046]: 0A07E19CE82: from=<vabazlov@client.vabazlov.net>, size=538, nrcpt=1 (queue active)
Nov 29 07:44:40 server postfix/smtp[14208]: 0A07E19CE82: to=<vabazlov@vabazlov.net>, relay=none, delay=0.01, delays=0/0.01/0/0, dsn=5.4.6, status
=bounced (mail for vabazlov.net loops back to myself)
Nov 29 07:44:40 server postfix/cleanup[14082]: 0DC1D19CE8A: message-id=<20251129074440.0DC1D19CE8A@server.vabazlov.net>
Nov 29 07:44:40 server postfix/bounce[14209]: 0A07E19CE82: sender non-delivery notification: 0DC1D19CE8A
Nov 29 07:44:40 server postfix/qmgr[14046]: 0DC1D19CE8A: from=<>, size=2510, nrcpt=1 (queue active)
Nov 29 07:44:40 server postfix/qmgr[14046]: 0A07E19CE82: removed
Nov 29 07:44:40 server postfix/smtp[14208]: connect to client.vabazlov.net[192.168.1.30]:25: No route to host
Nov 29 07:44:40 server postfix/smtp[14208]: 0DC1D19CE8A: to=<vabazlov@client.vabazlov.net>, relay=none, delay=0, delays=0/0/0/0, dsn=4.4.1, statu
s=deferred (connect to client.vabazlov.net[192.168.1.30]:25: No route to host)
```

Рис. 12: Журнал сервера

Open ▼ +

vabazlov.net
/var/named/master/fz

```
1 $ORIGIN .
2 $TTL 86400      ; 1 day
3 vabazlov.net    IN SOA  vabazlov.net. server.vabazlov.net. (
4                                2025112910 ; serial
5                                86400      ; refresh (1 day)
6                                3600       ; retry (1 hour)
7                                604800    ; expire (1 week)
8                                10800     ; minimum (3 hours)
9                                )
10                 NS      vabazlov.net.
11                 A       192.168.1.1
12                 MX 10   mail.vabazlov.net.
13 $ORIGIN vabazlov.net.
14 $TTL 1200      ; 20 minutes
15 client        A       192.168.1.30
16                 DHCID  ( AAEBuep8Zjsc9nlz8KLS80niDqaJEasdoIY5EwDQdIYS
17                 0Y0= ) ; 1 1 32
18 $TTL 86400      ; 1 day
19 dhcp          A       192.168.1.1
20 ns            A       192.168.1.1
21 server        A       192.168.1.1
22 www           A       192.168.1.1
23 mail          A       192.168.1.1
24
```



```
1 $ORIGIN .
2 $TTL 86400      ; 1 day
3 1.168.192.in-addr.arpa IN SOA 1.168.192.in-addr.arpa. server.vabazlov.net. (
4                               2025112907 ; serial
5                               86400      ; refresh (1 day)
6                               3600       ; retry (1 hour)
7                               604800    ; expire (1 week)
8                               10800     ; minimum (3 hours)
9                               )
10                          NS      1.168.192.in-addr.arpa.
11                          A       192.168.1.1
12                          PTR     server.vabazlov.net.
13                          MX 10   mail.vabazlov.net.
14 $ORIGIN 1.168.192.in-addr.arpa.
15 1                          PTR     server.vabazlov.net.
16                          PTR     ns.vabazlov.net.
17                          PTR     dhcp.vabazlov.net.
18                          PTR     www.vabazlov.net.
19                          PTR     mail.vabazlov.net.
20 $TTL 1200        ; 20 minutes
21 30                  PTR     client.vabazlov.net.
22                  DHCID    ( AAEBuep8Zjsc9nlz8KLS80niDqaJEasdoIY5EwDQdIYS
23                  0Y0= ) ; 1 1 32
```

```
[root@server.vabazlov.net ~]# cd /etc/namedb/master/ && cp /etc/namedb/master/masterfile.master /etc/namedb/master/masterfile.mydestination
[root@server.vabazlov.net ~]#
[root@server.vabazlov.net ~]# postconf -e 'mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain'
[root@server.vabazlov.net ~]# postfix check
[root@server.vabazlov.net ~]# systemctl reload postfix.service
[root@server.vabazlov.net ~]# restorecon -vR /etc
[root@server.vabazlov.net ~]# restorecon -vR /var/named/
[root@server.vabazlov.net ~]# systemctl restart postfix.service
[root@server.vabazlov.net ~]# systemctl restart named
[root@server.vabazlov.net ~]#
```

Рис. 15: Изменение mydestination

```
Nov 29 07:50:27 server postfix/smtpd[15341]: connect from client.vabazlov.net[192.168.1.30]
Nov 29 07:50:27 server postfix/smtpd[15341]: 69A7719B95C: client=client.vabazlov.net[192.168.1.30]
Nov 29 07:50:27 server postfix/cleanup[15344]: 69A7719B95C: message-id=<20251129075027.6181A6134533@client.vabazlov.net>
Nov 29 07:50:27 server postfix/smtpd[15341]: disconnect from client.vabazlov.net[192.168.1.30] ehlo=2 starttls=1 mail=1 rcpt=1 data=1 quit=1 commands=7
Nov 29 07:50:27 server postfix/qmgr[15251]: 69A7719B95C: from=<vabazlov@client.vabazlov.net>, size=538, nrcpt=1 (queue active)
Nov 29 07:50:27 server postfix/local[15345]: 69A7719B95C: to=<vabazlov@vabazlov.net>, relay=local, delay=0.01, delays=0/0/0/0, dsn=2.0.0, status=sent (delivered to mailbox)
Nov 29 07:50:27 server postfix/qmgr[15251]: 69A7719B95C: removed
```

Рис. 16: Успешная обработка письма сервером

Итоги работы

- Настроен сервер Postfix для локальной и доменной почты
- Реализована отправка между двумя узлами
- Исправлены ошибки DNS
- Добавлены MX и PTR записи
- Проверена доставка в логах
- Создан скрипт автоматизации окружения

Работа выполнена успешно.